Thaddeus Harris | Materials for the built environment | Homework #2 | 9/11/2024

(2) A 1 inch by 1 inch square steel bar is 12 inches long. The bar is loaded in tension so that it experiences an axial strain of 0.5%.

(a) At this strain, what are the new dimensions & cross sectional area of the bar if Poisson's Ratio is 0.3?

$\varepsilon_x = 0.005 = \frac{\Delta L}{12}$

$\Delta L_x = 0.005 \cdot 12$

$\Delta L_x = 0.06$

$L_x = \Delta L_x + L_{ox}$

$L_x = 0.06 + 12$

$L_x = 12.06$

$r = 0.3 = -\frac{\varepsilon_y}{\varepsilon_x} =$

$0.3 = -\frac{\varepsilon_y}{0.005}$

$-\varepsilon_y = 0.0015$

$\varepsilon_y = -0.0015$

$\Delta L_y = \varepsilon_y \cdot L_{oy}$

$\Delta L_y = -0.0015 \cdot 1$

$\Delta L_y = -0.0015$

$L_y = \Delta L_y + L_{oy}$

$L_y = -0.0015 + 1 = 0.9985$

$$\boxed{\begin{aligned} L_x &= 12.06 \\ L_y &= 0.9985 \end{aligned}}$$

(b) E = 29000 ksi for the steel. Based on the original bar dimensions and 0.5 percent axial strain what tensile stress exists in the bar assuming linear elastic behavior?

$E = \frac{\Delta\sigma}{\Delta\varepsilon} = 29000 \text{ ksi}$

$\frac{\Delta\sigma}{0.005} = 29000 \text{ ksi}$

$\Delta\sigma = 29000 \cdot 0.005$

$\sigma = 145 \text{ ksi}$

(c) What tensile force is required to produce the stress calculated in part (b)

$\sigma = 145 \text{ ksi} = \frac{F}{A}$

$145 \text{ ksi} = \frac{F}{1\text{in} \cdot 1\text{in}}$

$145 \text{ N/in}^2 = F/\text{in}^2$

$F = 145 \text{ k lb}$ lbs

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